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Paper Code : 21306

F–406

B. C. A. (Second Semester)
EXAMINATION, 2018

(New Course)

Paper No. BCA—N–201

DIGITAL ELECTRONICS

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions. All questions carry equal marks.

1. (a) Write the truth table and symbol of following gates :
   (i) XOR
   (ii) NAND
   (iii) NOR
   (b) Represent decimal number 27 in binary form using :
   (i) BCD Code
   (ii) Excess-3 Code
   (iii) Gray Code

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2. (a) Perform using 2's complement method :
   (i) 0011.1001 – 0001.1110
   (ii) 01100 – 00011
   (b) Minimize the four variable logic function using K-Map :

   $F(A, B, C, D) = \Sigma_m (0, 1, 2, 3, 5, 7, 8, 9, 11, 14)$

3. (a) Make a 32 : 1 MUX using 16 : 1 MUX.
   (b) Design a full adder circuit and realize it.
4. (a) What do you mean by Race around condition in JK Flip Flop ? How can it be solved ? Explain.
   (b) What is the purpose of counter ? Draw the logic diagram of 3-bit synchronous counter.
5. (a) State difference between Serial Registers and Parallel Registers.
   (b) Compare and contract state table and excitation table used in sequential circuits.
6. (a) Simplify the following expressions using Boolean Algebra :

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   (i) $AB + A(CD + \overline{CD'})$
   (ii) $\bar{A}'B + AB\bar{C}' + ABC$
   (b) Design a full subtractor circuit with three inputs *x*, *y* and *z* and two outputs D and B. The circuit subtracts the bits *x*-*y*-*z*, where *z* is the input borrow, B is the output borrow and D is the difference.

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7. Design a combinational circuit with three inputs, x, y and z, and the three outputs, A, B, and C. When the binary input is 0, 1, 2, or 3, the binary output is one greater than the input. When the binary input is 4, 5, 6, or 7 the binary output is one less than the input.

8. Write short notes on any *four* of the following :

   (a) SR Flip Flop

   (b) Decoder and Demultiplexer

   (c) BCD Ripple counter

   (d) De Morgan's theorem

   (e) SOP and POS to design combinational circuit

   (f) Floating point representation

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